

TECHNICAL PRESENTATION

Sai Kiran Reddy, Konda

Introduction → Education → Career → Technical Presentation → Q&A

Role: Engineer

Focus: Distributed Systems

Hardware foundations. Software ambitions.



Anna University
B.E. Electronics & Communication Engineering
Microprocessors, Embedded Systems, Digital Electronics



Northeastern University
M.S. Computer Software Engineering
Algorithms, Database Design, Cloud, DevOps



COMPANY	ROLE	HIGHLIGHTS
Cognizant	Project Lead	Mainframe → ETL migration. 4 teams, delivered through COVID.
Digital HIE	Tech Lead	HIPAA accreditation, solo-built compliance infrastructure.
Capital One	Systems Architect	130M+ accounts migrated to serverless AWS architecture.

One monolith. Four LOBs. It couldn't scale.

BEFORE		AFTER	
Architecture	Monolith	Architecture	Microservices
Coupling	Tight	Coupling	Loose
Deployment	All-or-nothing	Deployment	Independent
Scaling	Vertical only	Scaling	Horizontal
Risk	Shared failures	Risk	Isolated



Let the product decide the stack.

PHASE	DECISION
Requirements	130M+ accounts, 4 LOBs
Schema	NoSQL single-table design
Database	DynamoDB
Language	Go (concurrency + cold start)
Cloud	AWS Lambda + SQS + ALB

REQUEST FLOW



Automated pipeline. 98% coverage. Zero manual gates.

DEPLOYMENT PIPELINE



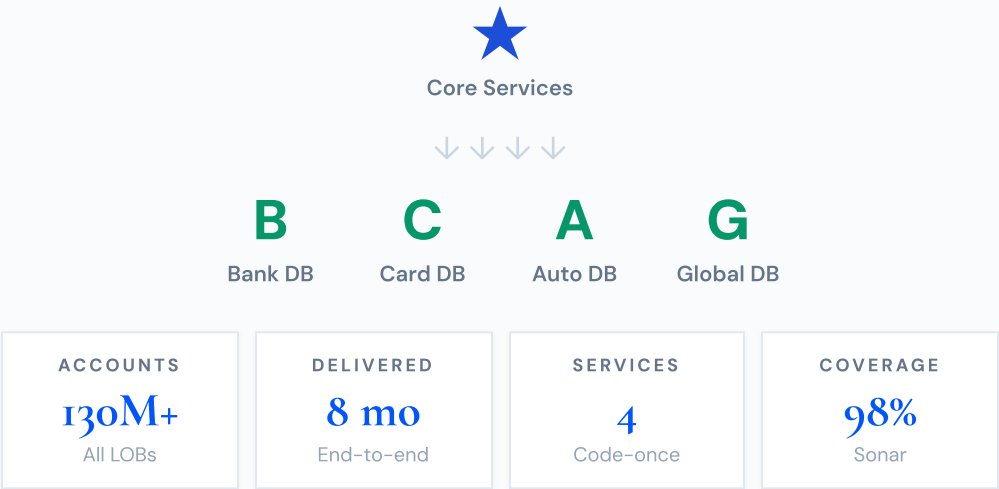
TEST TYPE	RESULT
SonarQube	98% Quality Gate
Unit Tests	100% Coverage
Integration	BDD / Gherkin
Performance	1B+ Records

BDD-first approach: Business stakeholders wrote Gherkin specs before engineering started. Every feature traced back to an acceptance criterion. Zero ambiguity at handoff.

Performance validated against 1B+ synthetic records across all four LOBs before production cutover.

Challenge existing designs. *Let the product decide.*

SERVICE ARCHITECTURE



- **Product requirements define architecture.** Schema → DB → Cloud → Language.
- **Challenge inherited designs.** Better approaches hide behind assumptions.
- **Root-cause analysis over gut.** Volumes + growth rate = foundation for every decision.
- **Ripple effect.** Other teams adopted the patterns including batch processing.

"Terraform-managed cross-region failover gave us active-active resilience without manual intervention. One config, two regions, zero downtime."

Beyond the code.



Sports

Volleyball, Cricket, Badminton.



Driving

Manhattan & Jersey City explorer.



Languages

English, Telugu, Tamil, Hindi, Kannada.



Exploring

New cuisines, new places, new conversations.

Ok, shoot.

Happy to go deeper on any decisions, trade-offs, or challenges.

[LinkedIn](#)

konda.saik@northeastern.edu